Corporate Finance

Lecture 7: Capital Structure in a Perfect Market (I)

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Plan

- Recap from lecture 6
- Ask whether financing a project with debt or with equity changes what it is worth
- Follow one entrepreneur who finances the same project two ways, and see why debt makes the equity riskier
- Meet Modigliani and Miller's first proposition, and the arbitrage behind it
- Read a market value balance sheet and follow a leveraged recapitalization

Mostly based on the BDM textbook Chap. 14.1-14.2.

Recap lecture 6

Recap of Lecture 6

Let's do a recap:

- The equity cost of capital
- The market risk premium
- Estimating beta
- The debt cost of capital
- A project's cost of capital: unlevering a comparable
- Cash as negative debt
- The weighted average cost of capital (WACC)

Recap: the equity cost of capital

The cost of capital of an investment is the expected return of the available investments with the same beta (Chap. 12.1). With the CAPM:

$$r_E = r_f + \beta_E \left(E[R_M] - r_f \right)$$

- r_f is the risk-free rate, $E[R_M] - r_f$ the market risk premium, and β_E how much the stock moves when the market moves by 1%.
- **Example.** Risk-free rate 3%, expected market return 8%. Saint-Gobain has a beta of 1.45: cost of equity $3\% + 1.45 \times 5\% = 10.25\%$. ENGIE has a beta of 0.82: $3\% + 0.82 \times 5\% = 7.1\%$.
- ENGIE is the more volatile stock (38% against 30%), yet its equity is cheaper. **Only market risk is paid**; the rest can be diversified away.

Recap: the market risk premium

The premium $E[R_M] - r_f$ cannot be observed, so it has to be estimated (Chap. 12.2).

- **Historical approach**: what the market actually paid over the risk-free rate in the past. Data-hungry, and backward looking.
- **Fundamental approach**: the return that justifies today's index level, given what firms are expected to pay out. With dividends growing at a constant rate g:

$$r_M = \frac{\text{Div}_1}{P_0} + g$$

  the dividend yield plus the growth rate.
- Same number seen from two ends: past realised returns, or expected future cash flows.

Recap: estimating beta

Beta is the slope of the best-fitting line through the stock's excess returns plotted against the market's excess returns (Chap. 12.3).

- **Example.** LVMH, monthly data: estimated beta 1.13, with a 95% confidence interval of [0.98; 1.27].
- With a risk-free rate of 3% and a market risk premium of 5%, the cost of equity lies between $3\% + 0.98 \times 5\% = 7.9\%$ and $3\% + 1.27 \times 5\% = 9.35\%$.
- A beta is an estimate, so the cost of capital it gives is a range, not a point.

Recap: the debt cost of capital

The yield to maturity of a bond overstates what lenders expect to earn when the borrower may default (Chap. 12.4):

$$r_D = \text{YTM} - p \times L$$

- p is the probability of default and L the expected loss rate when it happens. For a safe borrower p is close to zero and r_D is the yield.
- Alternatively, use the CAPM with a **debt beta**. Debt betas are small: close to zero for investment-grade bonds, higher for risky ones.
- **Example.** A BB-rated Lafarge bond with a yield of 3.7%, a 60% loss rate, and the default rate of its rating class: the expected return is well below the yield.

Recap: a project's cost of capital

To price a project, find a listed **comparable** with the same business risk, then **unlever** it: remove the effect of its borrowing (Chap. 12.5).

$$r_U = \frac{E}{E+D}\, r_E + \frac{D}{E+D}\, r_D \qquad \beta_U = \frac{E}{E+D}\, \beta_E + \frac{D}{E+D}\, \beta_D$$

- E and D are the market values of the comparable's equity and debt. The assets are owned by shareholders and lenders together, so their return is the **weighted average** of what each group earns. Same for the betas.
- **Example.** Danone: equity 30 billion, debt 10 billion, equity beta 0.4, debt yield 3.5% with negligible default risk. Cost of equity $3\% + 0.4 \times 5\% = 5\%$, so $r_U = 0.75 \times 5\% + 0.25 \times 3.5\% = 4.6\%$.
- **Borrowing does not make the business riskier.** It only decides who carries the risk: lenders take the safe part, shareholders the rest.

Recap: cash as negative debt

Cash that could be used to repay debt at once acts as **negative debt**: net debt = debt − cash.

- **Example.** Accenture, end of 2010: market capitalisation 25 billion, no debt, 4 billion in cash, equity beta 1.
- Net debt is $0 - 4 = -4$ billion, so the business itself (the enterprise value) is worth $25 - 4 = 21$ billion.
- The beta of the business is $\beta_U = \frac{25}{21} \times 1 = 1.19$: the cash made the shares **safer** than the business behind them.

Recap: the WACC

When a firm borrows, interest is tax deductible, so the effective cost of debt is $r_D(1-\tau_C)$ and the **weighted average cost of capital** is (Chap. 12.6)

$$r_{WACC} = \frac{E}{E+D}\, r_E + \frac{D}{E+D}\, r_D\,(1-\tau_C)$$

- τ_C is the corporate tax rate. Without the tax term the formula is the unlevered cost of capital r_U of the previous slide, also called the **pre-tax WACC**.
- **Example.** Michelon: equity 100 million, debt 25 million, cost of equity 10%, cost of debt 6%, tax rate 33%. Pre-tax WACC $0.8 \times 10\% + 0.2 \times 6\% = 9.2\%$; WACC $0.8 \times 10\% + 0.2 \times 6\% \times 0.67 = 8.8\%$.
- **Today we switch taxes off** and ask what borrowing does on its own.

Beginning of lecture 7 content

Lecture 6 measured the cost of equity and the cost of debt, and debt looked cheaper. Today we ask whether borrowing more therefore makes a firm worth more.

Introduction

- Dan Harris is the CFO of Electronic Business Services (EBS). The firm plans a big expansion and must **raise 50 million** from outside investors.
- Selling shares: given the firm's risk, equity investors want a 10% premium over the 5% risk-free rate. The **cost of equity is 15%**.
- Some executives object: EBS has never borrowed and has a strong balance sheet, so it could **borrow at 6%**. Why pay 15% when money is available at 6%?
- Three questions for Dan:
  1. Does the low interest rate make borrowing the better choice?
  2. Does the choice change the NPV of the expansion?
  3. Does it change the value of the firm, and its share price?

In a perfect capital market the answer is no, no and no. Today we see why. The next lectures see what changes in the real world.

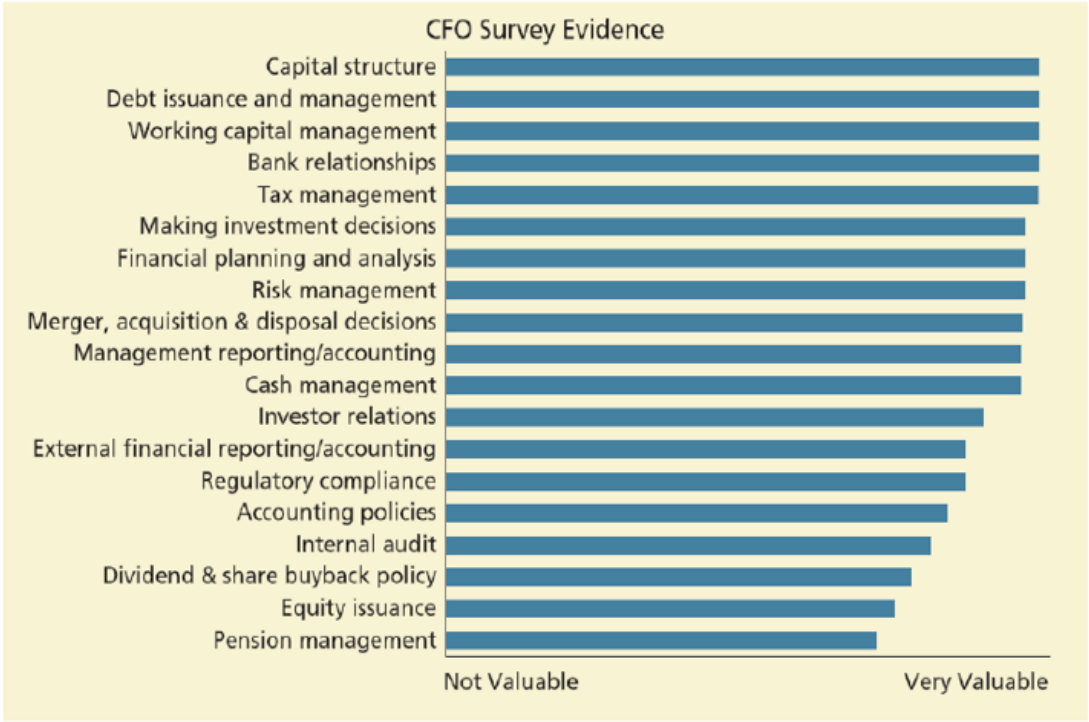
What CFOs think matters most

Figure 1. What finance functions add the most to firm value? (Source: Figure 1.1 in Graham, Smart, Megginson, 2010; original data from Tufano and Servaes, 2005)

FIGURE 1.1

Which Finance Functions Add the Most Value?

Source: Servaes and Tufano, "CFO Views on the Importance and Execution of the Finance Function" (Deutsche Bank, 2006).



- CFOs rank **capital structure** as the finance function that adds the most value. Modigliani and Miller say that in a perfect market it adds none. The gap between the two is the subject of Lectures 7 to 11.

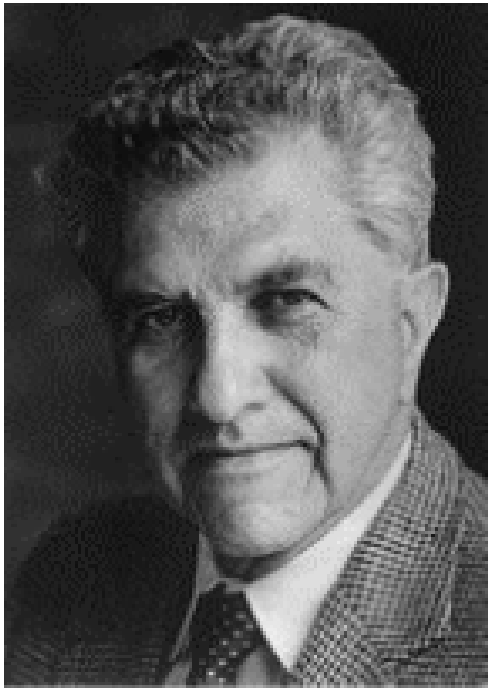
Not this Modigliani...



This Modigliani, and Miller



Franco Modigliani (1918-2003), Nobel prize 1985



Merton H. Miller (1923-2000), Nobel prize 1990

- "The Cost of Capital, Corporation Finance and the Theory of Investment," *American Economic Review*, 1958.
- Their two propositions tell us what leverage does to the value of the firm (today) and to its cost of capital (Lecture 8).

Outline

A. Financing a project with equity (Chap. 14.1)

B. Adding debt: levered equity and the 543 fallacy (Chap. 14.1)

C. Perfect markets and Modigliani-Miller Proposition I (Chap. 14.2)

D. Homemade leverage and arbitrage (Chap. 14.2)

E. The market value balance sheet (Chap. 14.2)

A. Financing with equity: the answer first

The answer first. The financing mix changes **who receives** the project's cash flows, not **how much** there is. In a perfect market the firm is worth the same with or without debt.

- An entrepreneur (call the firm Tour moche SA) has a project: invest **800** today; next year it pays **1400** if the economy is strong or **900** if it is weak. Both outcomes are equally likely.

Year 0: investment	Year 1: strong (prob. 0.5)	Year 1: weak (prob. 0.5)
-800	1400	900

- The cash flows depend on the economy, so they carry market risk. The risk-free rate is 5% and the appropriate risk premium is 10%: the project's **cost of capital is 15%**.
- Two steps: what is the project worth, and what do shareholders earn if the firm sells only equity?

Step 1: what is the project worth?

- Expected cash flow next year: $0.5 \times 1400 + 0.5 \times 900 = 1150$.
- Discounted at 15%:

$$\text{PV} = \frac{1150}{1.15} = 1000 \qquad \text{NPV} = -800 + 1000 = 200$$

- If the firm has no debt, shareholders receive all the cash flows. So the entrepreneur can sell the **whole equity for 1000**, pay the 800 investment, and **keep 200**: the NPV goes to the current owner.
- Equity in a firm with no debt is called **unlevered equity**.

Step 2: returns to unlevered equity

- The shares cost 1000 today and pay 1400 or 900 next year:

Unlevered equity	Year 0	Strong	Weak	Return strong	Return weak
Value	1000	1400	900	$+40\%$	-10%

- Expected return: $0.5 \times 40\% + 0.5 \times (-10\%) = 15\%$.
- The risk of the unlevered equity **is** the risk of the project, so 15% is exactly the fair return for the risk shareholders take.

With equity only: the firm is worth 1000, the shareholders expect 15%, the entrepreneur pockets 200.

B. Adding debt: the answer first

The answer first. Borrow 500 at 5% and the equity is worth $1000 - 500 = 500$. The firm as a whole is **still worth 1000**.

- The entrepreneur can also **borrow 500** and sell equity for the rest. The project pays at least 900, always enough to repay, so the debt is risk free and costs the risk-free rate: the firm will owe $500 \times 1.05 = 525$.
- Lenders are paid first. Shareholders get what is left: $1400 - 525 = 875$ or $900 - 525 = 375$. Equity in a firm that also has debt is **levered equity**.

Levered firm	Year 0	Strong	Weak
Debt	500	525	525
Levered equity	$E = ?$	875	375
Firm	?	1400	900

Step 1: the Law of One Price

- Add up what the lenders and the shareholders receive:
  - strong economy: $525 + 875 = 1400$
  - weak economy: $525 + 375 = 900$
- Together they receive **exactly the project's cash flows**. Two claims with the same cash flows must have the same price (Law of One Price), so

$$D + E = 1000 \qquad \Rightarrow \qquad E = 1000 - 500 = 500.$$

- The entrepreneur raises $500 + 500 = 1000$, just as before, and still keeps the NPV of 200.

Levered equity sells for less (500 against 1000) because it receives less. But the entrepreneur is not worse off: she raises the same 1000. She is **indifferent** between the two financings.

Step 2: the 543 fallacy

- Before MM, most people thought leverage **would** add value: levered equity pays $0.5 \times 875 + 0.5 \times 375 = 625$ on average, so

$$E = \frac{625}{1.15} = 543 > 500 \text{ ?!}$$

- **Where is the mistake?**

Security	Year 0	Strong	Weak	Return strong	Return weak	Expected
Debt	500	525	525	5%	5%	5%
Levered equity	500	875	375	$+75\%$	-25%	25%
Unlevered equity	1000	1400	900	$+40\%$	-10%	15%

- Levered equity swings $+75\%$ / -25%: **riskier**, so it requires 25%, and $625/1.25 = 500$.

Step 3: how much riskier?

- In a two-state world, a security's exposure to the economy is simply the gap between its return in the strong and in the weak state (the two-state version of beta):

Security	Return strong − return weak	Risk premium (expected − 5%)
Debt	$5\% - 5\% = 0\%$	0%
Unlevered equity	$40\% - (-10\%) = 50\%$	10%
Levered equity	$75\% - (-25\%) = 100\%$	20%

- Levered equity has **twice** the systematic risk of unlevered equity and earns **twice** the risk premium. The extra return is exactly the pay for the extra risk.
- Note that the debt is risk free: leverage raises the risk of equity **even when there is no risk of default**.

Step 4: debt is cheap, equity gets dearer

- Debt costs 5% and levered equity 25%. The average cost of the two sources, weighted by their values, is

$$\tfrac{500}{1000} \times 5\% + \tfrac{500}{1000} \times 25\% = 15\%,$$

  the same as with equity alone.

- **Example 14.1.** Borrow only 200. The firm owes 210; equity is worth 800 and receives $1400 - 210 = 1190$ or $900 - 210 = 690$: returns $+48.75\%$ or -13.75%, expected **17.5%**. Average cost: $0.8 \times 17.5\% + 0.2 \times 5\% = 15\%$ again.

Debt is cheaper when looked at on its own. But each unit of debt makes the equity riskier and dearer, by exactly enough to keep the average at 15%. Lecture 8 turns this into a formula.

C. Perfect markets and MM Proposition I

- Modigliani and Miller showed that the result holds in general, as long as capital markets are **perfect**:
  1. Investors and firms can trade the same securities at competitive prices equal to the present value of their cash flows.
  2. There are no taxes, transaction costs, or issuance costs.
  3. Financing decisions do not change the cash flows of the firm's investments, and do not reveal new information about them.

MM Proposition I. In a perfect capital market, the total value of a firm's securities is equal to the market value of the total cash flows generated by its assets, and is not affected by its choice of capital structure.

- In plain words: **the pie is the same size however you slice it.**

Why MM I holds: two ways to see it

- **The Law of One Price.** Without taxes or transaction costs, everything the firm's assets produce is paid to its security holders, and nothing else is. Same total cash flows, so same total value, whatever the mix of securities.
- **The separation principle.** When securities are fairly priced, buying or selling them is a **zero-NPV** transaction. The repayments a firm promises on its debt are worth exactly the loan it receives today. So borrowing creates no gain and no loss, and the value of the firm comes from its investments alone.
- Miller's version: the pizza delivery man asks Yogi Berra whether he wants his pizza cut into four or eight slices. Yogi: "Cut it in eight pieces, I'm feeling hungry tonight." Cutting the pizza differently does not make more pizza.

MM and the real world

- Students often ask why MM matters if real markets are not perfect.
- Galileo showed that **without friction**, all bodies fall at the same speed. Drop a feather and a stone and it does not hold. The reason is not that Galileo was wrong, but that **air friction** slows some objects more than others.
- MM is the same kind of result. In practice capital structure **does** affect value. MM tells us that any such effect must come from a **friction** in the capital market: taxes, bankruptcy costs, information problems.

MM tells us where to look: at the frictions, not at the mix itself. Lectures 9 to 11 are about those frictions.

D. Homemade leverage: the answer first

The answer first. Investors can borrow or lend on their own to **undo or redo** the firm's leverage. The firm's choice gives them nothing they cannot make themselves, so it cannot be worth anything to them.

- Suppose investors would prefer a different capital structure from the one the firm has chosen.
- An investor who wants more leverage can borrow and buy the shares: this is **homemade leverage**. One who wants less can hold the firm's debt as well as its shares.
- Condition: investors can borrow at the same rate as the firm. In a perfect market the rate depends only on the risk of the loan, not on who borrows.
- Three steps: make levered equity yourself, undo the firm's leverage yourself, and see what happens if the two firms are not priced the same.

Step 1: make levered equity yourself

- The firm has no debt. An investor who would like levered equity buys the unlevered shares with a **margin loan** of 500 at 5%:

Portfolio	Year 0: cost	Strong	Weak
Unlevered equity	1000	1400	900
Margin loan	−500	−525	−525
Total	500	875	375

- The shares are enough collateral, so the loan is risk free and costs 5%.
- The portfolio pays **exactly what levered equity pays**, for a cost of 500. By the Law of One Price, levered equity must also be worth 500.

Step 2: undo the firm's leverage

- Now the firm has borrowed 500, but the investor would prefer unlevered equity. He buys **both** the debt and the levered equity:

Portfolio	Year 0: cost	Strong	Weak
Debt	500	525	525
Levered equity	500	875	375
Total	1000	1400	900

- Same cash flows as unlevered equity, for 1000.

Either way, the investor picks the leverage he wants. The firm's choice changes nothing for him, so he will not pay more for one structure than for the other.

Step 3: if MM I failed, free money

- **Example 14.2.** Two firms with the same project (1400 or 900). Firm U has no debt and its equity trades at **990**. Firm L has borrowed 500 and its equity trades at **510**, so L is worth $510 + 500 = 1010$.

- Identical assets, different prices: the Law of One Price is violated. Build L's equity at home and sell the real one:

Trade	Year 0	Strong	Weak
Borrow 500	$+500$	-525	-525
Buy U's equity	-990	$+1400$	$+900$
Sell L's equity	$+510$	-875	-375
Total	$+20$	0	0

- **20 today, nothing to pay later**: an arbitrage. Everyone does it, the price of U rises and the price of L falls, until both firms are worth the same. That is what forces MM I to hold.

E. The market value balance sheet

The answer first. The market value of **all** the securities a firm has issued equals the market value of **all** its assets. At market values,

$$\text{Equity} = \text{Assets} - \text{Debt and other liabilities.}$$

Assets	Liabilities
Collection of Assets and Investments Undertaken by the Firm:	Collection of Securities Issued by the Firm:
Tangible Assets	Debt
Cash	Short-term debt
Plant, property, and equipment	Long-term debt
Inventory and other working capital (and so on)	Convertible debt
Intangible Assets	Equity
Intellectual property	Common stock
Reputation	Preferred stock
Human capital (and so on)	Warrants (options)
Total Market Value of Firm Assets	**Total Market Value of Firm Securities**

- Two differences with accounting: **every** asset counts (brand, people), all at **market value**. ATTENTION: single assets are often hard to value.

More than two securities

- MM I is not about debt and equity only. It holds for **any** set of securities, because investors can buy or sell them on their own.
- **Example 14.3.** The entrepreneur sells the firm as three securities: 500 of debt, equity, and a **warrant** that pays 210 if the economy is strong and nothing otherwise. The warrant is fairly priced at 60.
- Total value of the securities = value of the assets = 1000, so

$$E = 1000 - 500 - 60 = 440.$$

- The firm is worth 1000 whatever it issues. Adding a security only changes how the 1000 is divided.

A leveraged recapitalization

- MM I also applies **after** the firm is created. Harrison Industries is an all-equity firm with 50 million shares at 4 each: equity, and assets, are worth **200 million**.
- Harrison **borrows 80 million** and uses the cash to **buy back 20 million shares**: a leveraged recapitalization. Two stages, in millions:

Market value balance sheet	Initial	After borrowing	After repurchase
Cash	0	80	0
Existing assets	200	200	200
Total assets	**200**	**280**	**200**
Debt	0	80	80
Equity	200	200	120
Shares outstanding (M)	50	50	30
Value per share	4.00	4.00	4.00

Why the share price does not move

- **After borrowing**: assets and liabilities both rise by 80 (the cash in, the debt out). Equity is unchanged at 200, and so is the share price.
- **After the repurchase**: the 80 of cash is gone, the debt remains, so equity must fall by 80, to 120. But only $50 - 20 = 30$ million shares are left:

$$\frac{120}{30} = 4 \text{ per share, as before.}$$

Harrison sold 80 of debt and bought 80 of equity, both at fair prices. Benefits equal costs: a **zero-NPV** transaction, so it cannot change the price of the shares.

- Market capitalisation falls (200 to 120) because there are fewer shares, not because each share is worth less.

How it all connects

1. **The project is worth 1000** whatever the financing: its cash flows do not depend on who receives them.
2. **Debt gives equity a smaller but riskier slice**: 500 instead of 1000, swinging $+75\%$ / -25%.
3. **The 543 fallacy** is discounting the riskier slice at the unlevered rate. Twice the risk, twice the premium: 25%, not 15%.
4. **MM Proposition I**: in a perfect market the value of the firm does not depend on its capital structure, by the Law of One Price.
5. **Homemade leverage** makes the result bite: any gap between two identical firms is an arbitrage that investors close themselves.
6. **The market value balance sheet**: financing re-slices the value of the assets; a leveraged recap leaves the share price unchanged.

What to remember

- **MM Proposition I.** In a perfect capital market the total value of a firm's securities equals the value of its assets, whatever the capital structure.
- **Perfect market** means: securities trade at fair prices, no taxes or transaction costs, and financing changes neither the investment cash flows nor the information about them.
- **Leverage makes equity riskier**, even when the debt is risk free. Debt is cheaper on its own but raises the cost of equity.
- **Homemade leverage**: investors can borrow or lend to get the leverage they want, so the firm's choice is worth nothing to them.
- **Equity = assets − liabilities** at market values. Issuing debt to buy back shares at fair prices does not change the share price.

Next time

- In every example today the average cost of capital stayed at **15%**: debt cost 5%, equity became dearer, and the two effects cancelled.
- **Lecture 8** puts a formula on that: MM Proposition II tells us exactly how much the cost of equity rises with leverage, first in returns, then in betas.
- We will then use it to dismantle two popular arguments for debt: "leverage raises earnings per share" and "issuing equity dilutes shareholders".

Quiz

- Why is the levered equity of Tour moche SA worth 500 and not 543?
- Name the three conditions of a perfect capital market.
- The entrepreneur borrows 750 instead of 500. What is the value of the equity, what are its cash flows, and what is its expected return?
- A firm borrows to buy back shares in a perfect market. What happens to its share price? To its market capitalisation?

Quiz: answers

- **500, not 543.** The 543 discounts the expected levered cash flow of 625 at the unlevered rate of 15%. Levered equity is riskier and requires 25%: $625/1.25 = 500$. The 500 also follows from the Law of One Price: $1000 - 500$.
- **Perfect market.** Fair prices for all securities; no taxes, transaction or issuance costs; financing does not change the investment cash flows nor reveal information about them.
- **Borrow 750.** Equity is worth $1000 - 750 = 250$. The firm owes $750 \times 1.05 = 787.5$, so equity receives $1400 - 787.5 = 612.5$ or $900 - 787.5 = 112.5$: returns $+145\%$ or -55%, expected **45%**. Risk premium 40%, four times the unlevered premium, for four times the sensitivity.
- **Buyback.** The share price does not change (zero-NPV trade). Market capitalisation falls by the amount borrowed: fewer shares, same price.